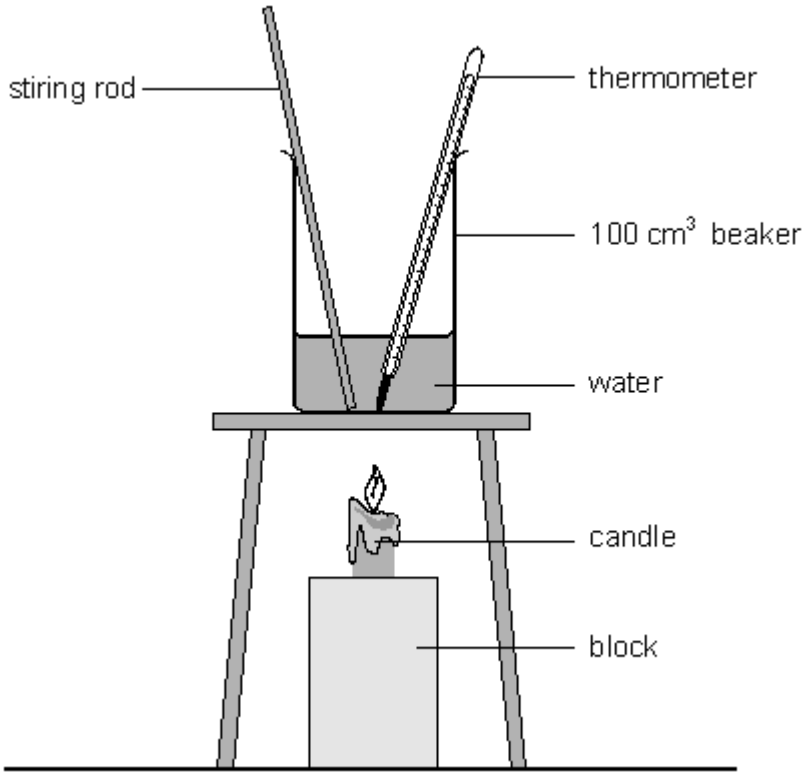
Q1. Luke investigated the heating of water. He predicted that the rise in temperature would depend on the volume of water.
The diagram shows the apparatus he used.



Luke recorded his results in a table as shown below.

beaker	volume of water, in cm^3	temperature at start, in °C	temperature after 2 minutes, in °C
A	25	18	30
B	50	18	24
C	75	18	22

(a) Why did Luke need to know the temperature of the water at the beginning and at the end of the experiment?

.....

.....

1 mark

(b) Did Luke's results support his prediction? Explain your answer.

.....

.....

1 mark

(c) Luke stirred the water during the experiment. How did this make his results more reliable?

.....

.....

1 mark

(d) Which of the following statements about the energy transferred to the beakers is correct? Tick the correct box.

Much more energy went into beaker 'A' because its temperature increased the most.	☐
The same amount of energy went into all three beakers.	☐
Beaker 'C' received the most energy because there was more water to heat.	☐

1 mark

(e) After a time, all three beakers cooled down to room temperature. What happened to the thermal energy in the beakers as they cooled down?

.....

.....

1 mark

Maximum 5 marks

Mark schemes

Q1.

(a) any **one** from

- to see how much the temperature went up
- to work out the temperature rise

*do **not** accept 'to make it a fair test'*

1 (L5)

(b) any **one** from

- yes because the smaller the volume the greater the rise in temperature
- yes because the greater the volume the smaller the rise in temperature

1 (L5)

(c) any **one** from

- it distributed the hot water throughout the beaker
- it made sure the water was heated evenly
- to make sure the temperature of the water was the same throughout

*do **not** accept 'the water heats up more quickly'*

1 (L5)

(d) The same amount of energy went into all three beakers. ✓

if more than one box is ticked, award no mark

1 (L6)

(e) any **one** from

- it decreased **or** got less

*accept 'it leaked out' **or** 'it was lost'*
accept 'it evaporated some of the water'

- it was transferred to the surroundings

accept 'it heated up the air'

1 (L6)

[5]